

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Blood Bank Management System (BBMS)

Project & Academic Information

- **Course:** Software Engineering Mini-Project
- **Institution:** Department of Computer Science & Engineering, PES University
- **Team Number:** T9
- **Project Allocation Sl. No.:** 9
- **Project Title:** Blood Bank Management System
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Team Members & Work Breakdown Structure (4-Way Module Split):

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Navaneeth Tanuboddi (Team Member 4 — Lead)	PES1UG24CS709 ([@Navaneeth1324](https://github.com/Navaneeth1324))	Module 4: ABO Compatibility Engine & UI	src/services/matchingEngine.js, src/services/requisitionService.js, src/routes/requisitionRoutes.js, src/public/* • Clinical ABO/Rh cross-matching & FEFO allocation engine • Hospital emergency requisition lifecycle & UI dashboard • Automated unit test suite & Cloud deployment

Document Revision History

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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to provide a complete, rigorous, and unambiguous description of the **Blood Bank Management System (BBMS)**. It details both functional and non-functional requirements, external interfaces, system behavior, data design, and architectural constraints. This document serves as the formal baseline for developers, testers, project evaluators, and academic faculty throughout the software engineering lifecycle.

1.2 Document Conventions

- This specification strictly follows the **IEEE Std 830-1998 Recommended Practice for Software Requirements Specifications**.
- Requirement identifiers use the notation `[FR-XX]` for Functional Requirements and `[NFR-XX]` for Non-Functional Requirements.
- Priority levels are defined using MoSCoW notation: **MUST**, **SHOULD**, **COULD**, or **WONT**.

1.3 Intended Audience and Reading Suggestions

- **Academic Evaluators / Faculty:** Read Sections 1, 2, 4, and 6 to evaluate engineering rigor, requirement specifications, and diagrammatic modeling.
- **System Developers:** Focus on Sections 3, 4, 6, and 7 for architectural, REST API, and relational database implementation details.
- **Quality Assurance & Testers:** Use Sections 4 and 5 to formulate test cases, acceptance criteria, and stress testing suites.

1.4 Product Scope

The **Blood Bank Management System (BBMS)** is an end-to-end, web-based healthcare logistics and inventory management solution. It bridges the critical communication gap between voluntary blood donors, blood banks, transfusion centers, and recipient hospitals.

Key capabilities include:

- Centralized tracking of blood units across 8 blood groups ($A^+ , A^- , B^+ , B^- , AB^+ , AB^- , O^+ , O^-$) and 4 blood components (Whole Blood, PRBC, Platelets, FFP).
- Automated donor eligibility screening (90-day cooldown enforcement, hemoglobin, and serological safety checks).
- Real-time stock visibility and shelf-life expiration alerts.
- Intelligent ABO/Rh compatibility cross-matching for emergency requisitions.

- Blood donation drive scheduling and emergency broadcast dispatch.

1.5 Definitions, Acronyms, and Abbreviations

- **BBMS**: Blood Bank Management System.
- **ABO System**: The classification of human blood based on the inherited properties of red blood cells (A, B, AB, O).
- **Rh Factor**: Rhesus factor (+ or -) denoting the presence or absence of Rh(D) antigen.
- **PRBC**: Packed Red Blood Cells (typical shelf-life: 42 days).
- **FFP**: Fresh Frozen Plasma (typical shelf-life: 1 year stored at -18°C).
- **Platelet Concentrate**: Thrombocyte component (typical shelf-life: 5 days at 20°C - 24°C with continuous agitation).
- **TTI**: Transfusion-Transmissible Infections (HIV, HBV, HCV, Syphilis, Malaria).
- **RBAC**: Role-Based Access Control.
- **FIFO / FEFO**: First In First Out / First Expired First Out allocation strategy.
- **DFD**: Data Flow Diagram.
- **ERD**: Entity Relationship Diagram.

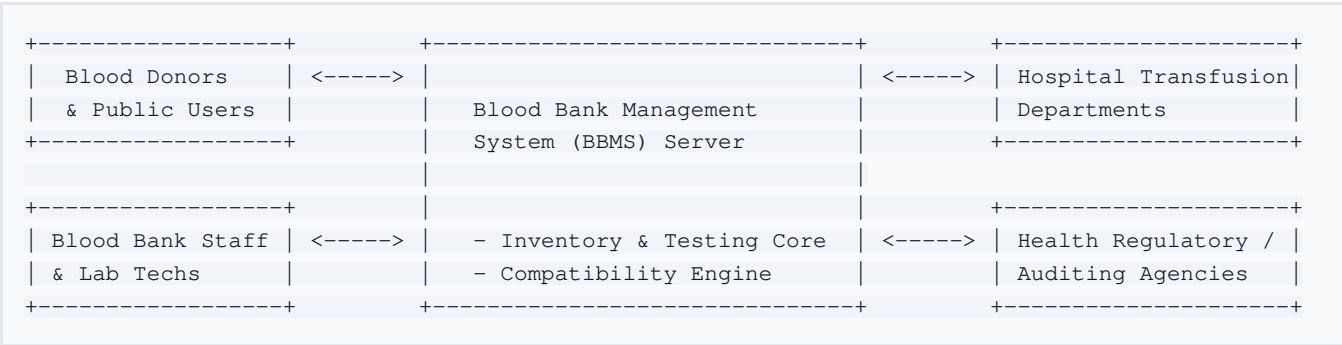
1.6 References

1. IEEE Std 830-1998, *IEEE Recommended Practice for Software Requirements Specifications*.
2. National Blood Transfusion Council (NBTC) & Drugs and Cosmetics Act (India) Guidelines for Blood Banking.
3. World Health Organization (WHO) Guidelines on Blood Donor Selection and Blood Inventory Management.
4. Pressman, R. S., *Software Engineering: A Practitioner's Approach*, McGraw-Hill.

2. Overall Description

2.1 Product Perspective

BBMS operates as a centralized web application running on modern client browsers and backed by a Node.js/Express application server with an embedded relational database engine (SQLite). It acts as an intermediary node in the healthcare network:



2.2 Product Functions

1. **Donor Registry & Health Profiling**: Tracks donor contact, medical history, physical vitals, and computes cooldown timelines.
2. **Component Separation & Inventory Ledger**: Generates unique bag barcode IDs, monitors component separation, shelf-life, and ambient storage racks.
3. **Hospital Requisition & Triage**: Receives requisitions, prioritizes based on emergency score (ROUTINE, URGENT, CRITICAL), and validates ABO compatibility.
4. **Blood Camp Organization**: Allows organizers to register camps, track attendee RSVPs, and register on-site

collections.

5. **Shortage Broadcast Alerts:** Triggers notification feeds when stock levels fall below safety thresholds.

6. **Regulatory Reporting:** Produces daily stock reports, discard/spoilage logs, and donor turnaround statistics.

2.3 User Classes and Characteristics

User Class	Technical Expertise	System Responsibilities	Access Privileges
System Administrator	High	System configuration, user account provisioning, audit log reviews, database backup.	Full administrative read/write access.
Blood Bank Staff / Lab Tech	Moderate	Screening donors, logging donations, entering TTI lab results, managing component stock, dispatching approved requests.	Operational inventory and testing read/write access.
Hospital Representative	Moderate	Submitting blood requests for admitted patients, tracking dispatch status, verifying receipt.	Requisition creation and tracking access.
Registered Donor	Low-to-Moderate	Viewing donation history, checking eligibility date, booking camp appointments, downloading donor certificate.	Self-profile and public search access.
Public / Guest Recipient	Low	Searching emergency blood availability by city/group, locating nearest blood banks.	Read-only public availability search.

2.4 Operating Environment

- **Server OS:** Linux (Ubuntu 22.04 LTS / Debian), macOS, Windows 10/11.
- **Runtime Environment:** Node.js (v18.x or higher LTS).
- **Client Platforms:** Modern desktop and mobile web browsers (Google Chrome 110+, Mozilla Firefox 110+, Safari 16+, Microsoft Edge).
- **Database:** SQLite3 (embedded) or PostgreSQL 14+.

2.5 Design and Implementation Constraints

- **Zero Configuration Portability:** The system must run smoothly in standard evaluation environments with an embedded zero-setup database.
- **Regulatory Strictness:** Blood units marked with reactive or unverified TTI results **MUST** be locked in quarantine and cannot be allocated.
- **Component Expiration Rigidity:** Units past their exact expiry timestamp **MUST** be automatically flagged as `EXPIRED` and forbidden from issuance.
- **FEFO Dispatch:** The compatibility engine **MUST** select compatible units using First-Expired-First-Out (FEFO) to minimize biological wastage.

2.6 Assumptions and Dependencies

- Donors provide accurate personal and medical screening answers.
- Blood testing for infectious diseases is performed on physical laboratory instruments, and technicians record binary pass/fail results into BBMS.
- Continuous network connectivity is available between hospital client terminals and the BBMS web server.

3. External Interface Requirements

3.1 User Interfaces

The system provides responsive web interfaces built with standard semantic HTML5, modern CSS3 (Tailwind styling), and dynamic JavaScript.

1. **Public Landing / Emergency Portal:** Prominent blood group search widget, real-time live availability indicators, and interactive camp listings.
2. **Staff / Admin Operations Console:** Tabular inventory matrix, batch blood intake form, lab test result verification modal, and requisition approval queue with FEFO suggestions.
3. **Hospital Portal:** Streamlined requisition form capturing Patient ID, Blood Group, Component, Quantity, Urgency Level, and Doctor's Prescription Ref.
4. **Donor Portal:** Visual eligibility meter (e.g., "Eligible in 14 days" or "Eligible to Donate Today"), donation timeline, and camp RSVP card.

3.2 Hardware Interfaces

- **Barcode / QR Scanner:** Standard HID (Human Interface Device) USB/Bluetooth scanners emitting text strings into input fields for bag barcode tracking.
- **Display Terminals:** Standard 1080p desktop monitors in blood bank facilities and mobile responsive viewports for donors.

3.3 Software Interfaces

- **Relational Database Management System:** SQLite3 relational database accessed via transactional SQL queries.
- **Cryptographic Libraries:** `bcryptjs` for salted password hashing; `jsonwebtoken` / cookie session management for stateless authentication.

3.4 Communications Interfaces

- **Protocol:** HTTP/1.1 and HTTPS using standard TLS 1.3 encryption.
- **Data Exchange Format:** JSON (JavaScript Object Notation) over RESTful APIs.

4. System Features & Functional Requirements

4.1 Module 1: Authentication & Role-Based Access Control (RBAC)

- **[FR-1.1] User Registration:** The system MUST allow donors and hospital representatives to register accounts with validated email, phone, and password credentials.
- **[FR-1.2] Secure Authentication:** The system MUST authenticate users using bcrypt-hashed passwords (minimum cost factor 10) and issue secure session tokens.
- **[FR-1.3] Role Authorization:** The system MUST restrict access to endpoints based on assigned roles (`ADMIN`, `STAFF`, `HOSPITAL`, `DONOR`).
- **[FR-1.4] Session Management:** The system MUST invalidate session tokens upon user logout and reject unauthorized requests with HTTP 401/403.

4.2 Module 2: Donor Registration & Medical Screening

- **[FR-2.1] Donor Profile Details:** The system MUST record donor full name, national ID/Aadhaar/SRN, date of birth, biological sex, contact number, address, and verified ABO/Rh blood group.
- **[FR-2.2] Vitals & Questionnaire Logging:** The system MUST record pre-donation vitals: weight (≥ 45 kg), hemoglobin (≥ 12.5 g/dL), systolic/diastolic blood pressure, and pulse.
- **[FR-2.3] 90-Day Cooldown Validation:** The system MUST compute the interval since the donor's last whole blood donation. If the interval is < 90 days, the system MUST reject donation scheduling with an eligibility error indicating days remaining.

- **[FR-2.4] Digital Donor Card:** The system MUST generate a personalized donor card displaying donor ID, blood group, total lifetime donations, and next eligible date.

4.3 Module 3: Blood Collection, Testing & Processing

- **[FR-3.1] Unique Barcode Generation:** For every donation, the system MUST generate a globally unique Blood Bag Unit Identifier (e.g., BLD-2026-001).
- **[FR-3.2] Serology / TTI Test Recording:** The system MUST require authorized staff to record results for HIV 1&2, Hepatitis B (HBsAg), Hepatitis C (HCV), Syphilis (VDRL), and Malaria.
- **[FR-3.3] Quarantine Locking:** Any unit with pending or reactive test results MUST remain in `TESTING` or `QUARANTINED` status and CANNOT be issued.
- **[FR-3.4] Component Separation:** The system MUST allow staff to record separation of a whole blood unit into derivatives:
 - Packed Red Blood Cells (PRBC) - Expiry: Collection Date + 42 Days.
 - Platelets - Expiry: Collection Date + 5 Days.
 - Fresh Frozen Plasma (FFP) - Expiry: Collection Date + 365 Days.

4.4 Module 4: Inventory Management & Expiry Tracking

- **[FR-4.1] Real-time Stock Matrix:** The system MUST maintain real-time counts across all 8 blood groups and 4 components.
- **[FR-4.2] FEFO (First Expired First Out) Sorting:** Stock queries for allocation MUST sort available units by `expiry_date ASC`.
- **[FR-4.3] Automated Expiry Flagging:** Units where `current_timestamp >= expiry_date` MUST automatically transition to `EXPIRED` status.
- **[FR-4.4] Discard Audit Logging:** The system MUST record reason, timestamp, and technician ID whenever an expired or contaminated unit is discarded.
- **[FR-4.5] Low Stock Thresholds:** The system MUST trigger visual alerts when any blood group stock falls below 5 units.

4.5 Module 5: Blood Requisition & Compatibility Matching

- **[FR-5.1] Hospital Request Submission:** The system MUST allow authenticated hospital representatives to submit requisitions with Patient Name, Hospital File Number, Required Blood Group, Component Type, Units Needed, and Urgency (`ROUTINE`, `URGENT`, `CRITICAL`).
- **[FR-5.2] ABO/Rh Compatibility Engine:** The system MUST apply clinical cross-match rules:
 - For Whole Blood / PRBC:
 - $\$O^-$ is universal donor (can be given to any group).
 - $\$O^+$ can be given to $\$O^+$, A^+ , B^+ , AB^+ .
 - $\$A^-$ can be given to $\$A^-$, A^+ , AB^- , AB^+ .
 - $\$A^+$ can be given to $\$A^+$, AB^+ .
 - $\$B^-$ can be given to $\$B^-$, B^+ , AB^- , AB^+ .
 - $\$B^+$ can be given to $\$B^+$, AB^+ .
 - $\$AB^-$ can be given to $\$AB^-$, AB^+ .
 - $\$AB^+$ can only be given to $\$AB^+$ (Universal recipient).
 - For Plasma (FFP):
 - $\$AB$ is the universal plasma donor; $\$O$ is universal plasma recipient.
- **[FR-5.3] Allocation & Reservation:** When a requisition is approved, the specified units MUST be marked `RESERVED` immediately to prevent race conditions.
- **[FR-5.4] Dispatch Confirmation:** Upon physical pickup, staff MUST mark the request `FULFILLED`, marking the

reserved bags as `DISPATCHED`.

4.6 Module 6: Blood Donation Camps & Drive Management

- **[FR-6.1] Camp Scheduling:** Admin/Staff MUST be able to create donation camps with Name, Venue, Date, Start/End Time, and Organizer Details.
- **[FR-6.2] Donor RSVP:** Registered donors MUST be able to pre-register for upcoming camps.
- **[FR-6.3] Drive Metrics:** The system MUST track total registrations and total units collected per camp.

4.7 Module 7: Emergency Alert & Broadcast System

- **[FR-7.1] Critical Shortage Trigger:** When critical requisitions cannot be fulfilled from stock, the system MUST generate an emergency broadcast notice.
- **[FR-7.2] Donor Notification Feed:** Donors with matching blood groups who are eligible (cooldown satisfied) MUST be surfaced for outreach.

4.8 Module 8: Analytics, Reports & Audit Trails

- **[FR-8.1] Executive Dashboard Metrics:** The system MUST display total donations, active inventory, pending requests, and upcoming expirations.
- **[FR-8.2] Audit Trail:** All state changes to blood units and requisitions MUST record actor ID, timestamp, and previous/new status.

5. Non-Functional Requirements

5.1 Performance Requirements

- **[NFR-1.1] Search Latency:** Blood stock availability queries MUST return within $\leq 200\text{ ms}$ under normal operating conditions.
- **[NFR-1.2] Concurrent Users:** The server architecture MUST support at least 50 concurrent transactions without deadlocks or race conditions during blood reservation.
- **[NFR-1.3] Page Load Time:** Frontend UI views MUST achieve full interactivity within $\leq 1.5\text{ seconds}$ on standard broadband connections.

5.2 Safety Requirements

- **[NFR-2.1] Biological Safety Guarantee:** Under no circumstances shall the system allow a unit with reactive serology or an expired timestamp to be marked as `AVAILABLE` or `DISPATCHED`.
- **[NFR-2.2] Accidental Deletion Prevention:** Hard deletion of blood unit records and completed requisitions is strictly prohibited; soft-deletions and audit logs MUST be enforced.

5.3 Security Requirements

- **[NFR-3.1] Password Cryptography:** Passwords MUST never be stored in plaintext. They MUST be hashed using `bcrypt` with salt rounds ≥ 10 .
- **[NFR-3.2] Input Sanitization & SQLi Defense:** All user inputs MUST be sanitized and executed via parameterized queries to eliminate SQL injection vulnerabilities.
- **[NFR-3.3] Cross-Site Scripting (XSS) Prevention:** All dynamic HTML output MUST be escaped properly.
- **[NFR-3.4] Data Confidentiality:** Sensitive donor medical history and test results MUST only be visible to authenticated medical staff and the donor themselves.

5.4 Software Quality Attributes

- **[NFR-4.1] Usability:** Intuitive user interface with clear visual hierarchy, color-coded status badges, and mobile-responsive layout.

- **[NFR-4.2] Reliability (MTBF):** The system MUST maintain 99.5% uptime during operational hours.
- **[NFR-4.3] Maintainability:** Modular codebase adhering to MVC/REST separation of concerns with clean documentation and seed routines.

6. System Models & Diagrams

6.1 Use Case Diagram

```
flowchart TD
    subgraph Actors
        D["Donor"]
        H["Hospital Representative"]
        S["Blood Bank Staff / Lab Tech"]
        A["System Admin"]
        P["Public / Patient"]
    end

    subgraph BBMS["Blood Bank Management System"]
        UC1["Register / Login (RBAC)"]
        UC2["Search Blood Availability"]
        UC3["Check Donation Eligibility (90d)"]
        UC4["Book Camp Appointment"]
        UC5["View Donor Card & History"]
        UC6["Submit Blood Requisition"]
        UC7["Track Requisition Status"]
        UC8["Screen Donor & Log Vitals"]
        UC9["Log Blood Collection & Barcode"]
        UC10["Record Serology / TTI Tests"]
        UC11["Manage Component Separation"]
        UC12["Monitor Stock & Expiry (FEFO)"]
        UC13["Match Compatibility & Approve Request"]
        UC14["Dispatch Blood Bags"]
        UC15["Schedule Blood Camps"]
        UC16["Manage Users & View Audit Logs"]
    end

    P --> UC2
    D --> UC1
    D --> UC2
    D --> UC3
    D --> UC4
    D --> UC5

    H --> UC1
    H --> UC2
    H --> UC6
    H --> UC7

    S --> UC1
    S --> UC8
    S --> UC9
    S --> UC10
    S --> UC11
    S --> UC12
    S --> UC13
    S --> UC14
    S --> UC15
```

```
A --> UC1
A --> UC16
A --> UC12
```

6.2 Data Flow Diagram (DFD) - Level 0 (Context Diagram)

```
flowchart LR
    Donor["Donor"]
    Hospital["Hospital"]
    Staff["Blood Bank Staff"]
    Public["Public / Patient"]

    BBMS(("0.0<br/>Blood Bank<br/>Management<br/>System (BBMS)"))

    Donor -->|"Donor Details, Health Vitals, Camp RSVP"| BBMS
    BBMS -->|"Eligibility Status, Donor Card, Camp Details"| Donor

    Hospital -->|"Blood Requisition, Patient Info, Urgency"| BBMS
    BBMS -->|"Requisition Status, Allocated Bag Barcodes"| Hospital

    Staff -->|"TTI Test Results, Component Separation, Bag Intake"| BBMS
    BBMS -->|"Stock Alerts, Expiry Warnings, Matching Suggestions"| Staff

    Public -->|"Availability Query (Blood Group, City)"| BBMS
    BBMS -->|"Real-time Stock Availability Counts"| Public
```

6.3 Data Flow Diagram (DFD) - Level 1 (Functional Decomposition)

```
flowchart TD
    D["Donor"]
    H["Hospital"]
    S["Staff / Lab Tech"]

    subgraph Processes
        P1(("1.0<br/>User & Donor<br/>Management"))
        P2(("2.0<br/>Testing &<br/>Component Lab"))
        P3(("3.0<br/>Inventory &<br/>Expiry Engine"))
        P4(("4.0<br/>Compatibility &<br/>Requisition Engine"))
        P5(("5.0<br/>Camp & Alert<br/>Coordination"))
    end

    subgraph DataStores["Data Stores"]
        DS1["D1: Users & Donors"]
        DS2["D2: Blood Units & Tests"]
        DS3["D3: Requisitions & Allocations"]
        DS4["D4: Camps & RSVPs"]
    end

    D -->|"Registration, Vitals"| P1
    P1 <-->|"Read/Write Profile & Cooldown"| DS1
    P1 -->|"Eligibility Confirmation"| D

    S -->|"Donation Intake, TTI Results"| P2
    P2 <-->|"Verify Donor"| DS1
    P2 -->|"Create Unit, Log Tests"| DS2

    P2 -->|"Cleared Stock"| P3
```

```

P3 <-->|"Update Stock, Expiry Audit"| DS2
P3 -->|"Low Stock Alerts"| S

H -->|"Blood Request"| P4
P4 <-->|"Check Stock & FEFO Match"| DS2
P4 <-->|"Write Request Record"| DS3
P4 -->|"Allocation Notification"| H
S -->|"Approve & Dispatch"| P4

S -->|"Schedule Camps"| P5
D -->|"RSVP"| P5
P5 <-->|"Camp Data"| DS4

```

6.4 Data Flow Diagram (DFD) - Level 2 (Requisition & Cross-Match Flow)

```

flowchart TD
    H["Hospital Representative"]
    Staff["Blood Bank Staff"]

    subgraph P4["Process 4.0: Requisition & Allocation Deep-Dive"]
        P41(("4.1<br/>Validate Request<br/>& Urgency"))
        P42(("4.2<br/>Query ABO/Rh<br/>Compatibility"))
        P43(("4.3<br/>FEFO Priority<br/>Selection"))
        P44(("4.4<br/>Reserve Bags<br/>& Prevent Race"))
        P45(("4.5<br/>Authorize &<br/>Dispatch"))
    end

    end

    DS_Req[("D3: Requisitions")]
    DS_Stock[("D2: Blood Units (Available)")]

    H -->|"1. Submit Request Details"| P41
    P41 -->|"2. Record Pending Request"| DS_Req
    P41 -->|"3. Trigger Match"| P42
    P42 <-->|"4. Filter by ABO Rules"| DS_Stock
    P42 -->|"5. Compatible Candidates"| P43
    P43 <-->|"6. Sort by Expiry Date ASC"| DS_Stock
    P43 -->|"7. Best Matching Units"| P44
    P44 -->|"8. Status -> RESERVED"| DS_Stock
    P44 -->|"9. Ready for Staff Review"| Staff
    Staff -->|"10. Staff Approves Pickup"| P45
    P45 -->|"11. Status -> DISPATCHED"| DS_Stock
    P45 -->|"12. Status -> FULFILLED"| DS_Req
    P45 -->|"13. Handover Slip"| H

```

6.5 Entity-Relationship (ER) Diagram

```

erDiagram
    USERS ||--o{ DONORS : "extends (if role=DONOR)"
    USERS ||--o{ HOSPITALS : "extends (if role=HOSPITAL)"
    DONORS ||--o{ DONATIONS : "makes"
    DONATIONS ||--|| BLOOD_UNITS : "yields"
    BLOOD_UNITS ||--o{ BLOOD_TESTS : "undergoes"
    HOSPITALS ||--o{ BLOOD_REQUESTS : "places"
    BLOOD_REQUESTS ||--o{ REQUEST_ALLOCATIONS : "allocates"
    BLOOD_UNITS ||--o{ REQUEST_ALLOCATIONS : "assigned_to"
    USERS ||--o{ BLOOD_CAMPS : "organizes"
    DONORS ||--o{ CAMP_REGISTRATIONS : "attends"

```

```
BLOOD_CAMPS ||--o{ CAMP_REGISTRATIONS : "hosts"
```

```
USERS {  
    int id PK  
    string email UK  
    string password_hash  
    string role "ADMIN | STAFF | DONOR | HOSPITAL"  
    string full_name  
    string phone  
    datetime created_at  
}
```

```
DONORS {  
    int id PK  
    int user_id FK  
    string blood_group "A+ | A- | B+ | B- | AB+ | AB- | O+ | O-"  
    date date_of_birth  
    string gender "M | F | Other"  
    float weight_kg  
    float hemoglobin  
    date last_donation_date  
    string medical_history  
}
```

```
HOSPITALS {  
    int id PK  
    int user_id FK  
    string hospital_name  
    string license_number  
    string address  
    string city  
    string emergency_contact  
}
```

```
DONATIONS {  
    int id PK  
    int donor_id FK  
    int camp_id FK "Nullable"  
    datetime donation_date  
    float systolic_bp  
    float diastolic_bp  
    float pulse_rate  
    string collected_by_staff  
}
```

```
BLOOD_UNITS {  
    int id PK  
    int donation_id FK  
    string barcode_id UK  
    string blood_group  
    string component_type "WHOLE_BLOOD | PRBC | PLATELETS | FFP"  
    int volume_ml  
    date collection_date  
    date expiry_date  
    string storage_rack  
    string status "TESTING | AVAILABLE | RESERVED | EXPIRED | DISPATCHED | DISCARDED"  
}
```

```
BLOOD_TESTS {  
    int id PK  
    int blood_unit_id FK  
    string test_name "HIV | HBV | HCV | SYPHILIS | MALARIA"
```

```

        string result "NEGATIVE | POSITIVE | PENDING"
        datetime tested_at
        string tested_by
    }

    BLOOD_REQUESTS {
        int id PK
        int hospital_id FK
        string patient_name
        string patient_id
        string blood_group
        string component_type
        int units_requested
        string urgency "ROUTINE | URGENT | CRITICAL"
        string doctor_name
        string status "PENDING | APPROVED | REJECTED | FULFILLED"
        datetime created_at
    }

    REQUEST_ALLOCATIONS {
        int id PK
        int request_id FK
        int blood_unit_id FK
        datetime allocated_at
    }

    BLOOD_CAMPS {
        int id PK
        string camp_name
        string venue
        date camp_date
        time start_time
        time end_time
        string organizer_contact
        int created_by FK
    }

    CAMP_REGISTRATIONS {
        int id PK
        int camp_id FK
        int donor_id FK
        datetime registered_at
        string status "REGISTERED | ATTENDED | CANCELLED"
    }

```

6.6 Sequence Diagram: Donor Registration & Blood Donation

```

sequenceDiagram
    autonumber
    actor Donor
    participant WebUI as Web Client (Frontend)
    participant DonorCtrl as Donor Controller
    participant LabCtrl as Lab & Inventory Controller
    participant DB as Relational Database

    Donor->>WebUI: Navigate to Register / Profile
    Donor->>WebUI: Submit Vitals (Weight, Hb, Blood Group)
    WebUI->>DonorCtrl: POST /api/donors/register
    DonorCtrl->>DB: Query last_donation_date for Donor

```

```

DB-->>DonorCtrl: Return last_donation_date
DonorCtrl->>DonorCtrl: Verify (today - last_donation >= 90 days) & weight >= 45 & Hb >= 12.5
alt Criteria Failed (Cooldown < 90 days)
    DonorCtrl-->>WebUI: HTTP 400: Not eligible (X days remaining)
    WebUI-->>Donor: Display Ineligibility Badge & Cooldown
else Criteria Passed
    DonorCtrl->>DB: INSERT into DONORS record
    DB-->>DonorCtrl: Success
    DonorCtrl-->>WebUI: HTTP 201: Registration & Appointment Confirmed
    WebUI-->>Donor: Render Digital Donor Card

    Note over Donor,LabCtrl: Physical Blood Donation at Center / Camp
    actor Staff
    Staff->>WebUI: Initiate Donation Intake
    WebUI->>LabCtrl: POST /api/inventory/intake (Donor ID, Bag Barcode, Volume)
    LabCtrl->>DB: INSERT into DONATIONS & BLOOD_UNITS (status='TESTING')
    DB-->>LabCtrl: Unit Created
    Staff->>WebUI: Enter Serology Results (HIV: Neg, HBV: Neg, HCV: Neg, etc.)
    WebUI->>LabCtrl: POST /api/inventory/tests (Bag ID, Results)
    LabCtrl->>LabCtrl: Validate all tests == 'NEGATIVE'
    LabCtrl->>DB: UPDATE BLOOD_UNITS status='AVAILABLE'
    DB-->>LabCtrl: Stock Updated
    LabCtrl-->>WebUI: Unit cleared for transfusion
end

```

6.7 Sequence Diagram: Hospital Emergency Requisition & Cross-Match

```

sequenceDiagram
    autonumber
    actor Hospital as Hospital Representative
    participant WebUI as Web Client
    participant ReqCtrl as Requisition Controller
    participant Engine as Compatibility & FEFO Engine
    participant DB as Relational Database
    actor Staff as Blood Bank Staff

    Hospital->>WebUI: Fill Emergency Requisition (Patient, Blood Group, Component, Urgency)
    WebUI->>ReqCtrl: POST /api/requests
    ReqCtrl->>DB: INSERT into BLOOD_REQUESTS (status='PENDING')
    DB-->>ReqCtrl: Request ID: REQ-1049
    ReqCtrl-->>WebUI: HTTP 201: Requisition Logged

    Staff->>WebUI: Open Requisition Queue
    WebUI->>ReqCtrl: GET /api/requests/pending
    ReqCtrl->>Engine: MatchCompatibleUnits(BloodGroup, Component, Quantity)
    Engine->>DB: SELECT * FROM BLOOD_UNITS WHERE status='AVAILABLE' AND expiry_date > NOW()
    DB-->>Engine: List of candidate bags
    Engine->>Engine: Filter by ABO Compatibility Matrix
    Engine->>Engine: Sort by FEFO (expiry_date ASC)
    Engine-->>ReqCtrl: Return Recommended Bag IDs
    ReqCtrl-->>WebUI: Display Request with FEFO Matching Suggestions

    Staff->>WebUI: Click "Approve & Reserve"
    WebUI->>ReqCtrl: POST /api/requests/:id/approve
    ReqCtrl->>DB: BEGIN TRANSACTION
    ReqCtrl->>DB: UPDATE BLOOD_UNITS SET status='RESERVED' WHERE id IN (...)
    ReqCtrl->>DB: INSERT into REQUEST_ALLOCATIONS
    ReqCtrl->>DB: UPDATE BLOOD_REQUESTS SET status='APPROVED'
    ReqCtrl->>DB: COMMIT TRANSACTION

```

```

DB-->>ReqCtrl: Transaction Success
ReqCtrl-->>WebUI: Approval Confirmed

Hospital->>Staff: Hospital Courier Arrives for Pickup
Staff->>WebUI: Confirm Handover / Dispatch
WebUI->>ReqCtrl: POST /api/requests/:id/dispatch
ReqCtrl->>DB: UPDATE BLOOD_UNITS SET status='DISPATCHED'
ReqCtrl->>DB: UPDATE BLOOD_REQUESTS SET status='FULFILLED'
DB-->>ReqCtrl: Updated
ReqCtrl-->>WebUI: Dispatch Receipt Generated

```

6.8 State Transition Diagram: Blood Unit Lifecycle

```

stateDiagram-v2
    [*] --> DONATED: Phlebotomy Collection
    DONATED --> TESTING: Transferred to Serology Lab

    state TESTING {
        [*] --> SCREENING
        SCREENING --> TESTS_PASSED: HIV, HBV, HCV, Syphilis, Malaria Negative
        SCREENING --> TESTS_REACTIVE: Any Positive TTI Result
    }

    TESTS_REACTIVE --> DISCARDED: Incinerated / Biohazard Protocol
    TESTS_PASSED --> SEPARATED: Component Processing (PRBC / FFP / Platelets)
    SEPARATED --> AVAILABLE: Barcoded & Stored in Refrigerated Rack

    AVAILABLE --> RESERVED: Matched with Approved Hospital Request
    AVAILABLE --> EXPIRED: Exceeded Shelf-Life (e.g., >42d for PRBC, >5d for Platelets)

    RESERVED --> DISPATCHED: Handed over to Hospital Transfusion Team
    RESERVED --> AVAILABLE: Request Cancelled by Hospital

    EXPIRED --> DISCARDED: Discard Log Recorded
    DISPATCHED --> [*]: Transfused to Patient
    DISCARDED --> [*]: Hazardous Waste Completed

```

6.9 System Architecture Diagram

```

flowchart TD
    subgraph ClientLayer["Client Layer (Presentation Tier)"]
        B1["Public Portal<br/>(Guest Search & Camps)"]
        B2["Donor Dashboard<br/>(Profile, Eligibility, RSVP)"]
        B3["Hospital Portal<br/>(Requisitions & Tracking)"]
        B4["Staff / Admin Console<br/>(Inventory, Testing, Approvals)"]
    end

    subgraph APIGateway["Application & Security Gateway"]
        AuthMiddleware["JWT / Session Auth & RBAC Guard"]
        RateLimiter["Rate Limiting & Input Validation"]
    end

    subgraph ServiceLayer["Business Logic & Service Layer (Node.js / Express)"]
        AuthService["Auth Service<br/>(bcrypt, roles)"]
        DonorService["Donor Service<br/>(90d cooldown, vitals)"]
        InventoryService["Inventory Service<br/>(Component, Expiry Tracker)"]
        CompatEngine["Cross-Match Engine<br/>(ABO/Rh Matrix, FEFO)"]
    end

```

```
    RequisitionService["Requisition Service<br/>(Allocation, Dispatch)"]
    CampService["Camp & Alert Service<br/>(Drives, Shortage Feeds)"]
end

subgraph DataAccessLayer["Data Persistence Layer"]
    ORM["Data Access / Query Controller"]
    DB[(Relational DB<br/>SQLite3)]
end

ClientLayer -->|"REST API / JSON"| APIGateway
APIGateway --> ServiceLayer
ServiceLayer --> DataAccessLayer
DataAccessLayer --> DB
```

7. Database Schema & Data Dictionary

Table 1: `users`

Column Name	Data Type	Constraints	Description
id	INTEGER	PRIMARY KEY AUTOINCREMENT	Unique system user ID
email	VARCHAR(120)	UNIQUE, NOT NULL	Login email address
password_hash	VARCHAR(255)	NOT NULL	Salted bcrypt hash of user password
role	VARCHAR(20)	NOT NULL	ADMIN, STAFF, HOSPITAL, or DONOR
full_name	VARCHAR(100)	NOT NULL	Name of individual or entity
phone	VARCHAR(20)	NOT NULL	Primary contact phone number
created_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	Record creation timestamp

Table 2: `donors`

Column Name	Data Type	Constraints	Description
id	INTEGER	PRIMARY KEY AUTOINCREMENT	Unique donor profile ID
user_id	INTEGER	FOREIGN KEY (users.id)	Linked authentication account
blood_group	VARCHAR(5)	NOT NULL	\$A^+, A^-, B^+, B^-, AB^+, AB^-, O^+, O^- \$
date_of_birth	DATE	NOT NULL	Used to verify age \$ge 18\$ years
gender	VARCHAR(10)	NOT NULL	Male, Female, or Other
weight_kg	DECIMAL(4,1)	NOT NULL	Must be \$ge 45.0\text{ kg}\$
hemoglobin	DECIMAL(3,1)	NOT NULL	Must be \$ge 12.5\text{ g/dL}\$
last_donation_date	DATE	NULL	Date of previous donation
medical_history	TEXT	NULL	Past medical conditions, medications

Table 3: `blood\_units`

Column Name	Data Type	Constraints	Description
id	INTEGER	PRIMARY KEY AUTOINCREMENT	Internal unit key
barcode_id	VARCHAR(50)	UNIQUE, NOT NULL	Bag barcode string (e.g. BLD-2026-001)
blood_group	VARCHAR(5)	NOT NULL	ABO/Rh group
component_type	VARCHAR(30)	NOT NULL	WHOLE_BLOOD, PRBC, PLATELETS, FFP
volume_ml	INTEGER	NOT NULL	Volume in milliliters (typically 350 or 450)
collection_date	DATE	NOT NULL	Date unit was collected
expiry_date	DATE	NOT NULL	Computed date of expiration
storage_rack	VARCHAR(30)	NOT NULL	Physical storage rack/shelf ID
status	VARCHAR(20)	NOT NULL	TESTING, AVAILABLE, RESERVED, EXPIRED, DISPATCHED, DISCARDED
donor_id	INTEGER	FOREIGN KEY (donors.id)	Originating donor

Table 4: `blood\_tests`

Column Name	Data Type	Constraints	Description
id	INTEGER	PRIMARY KEY AUTOINCREMENT	Test record key
blood_unit_id	INTEGER	FOREIGN KEY (blood_units.id)	Tested unit
hiv	VARCHAR(15)	NOT NULL	NEGATIVE, POSITIVE, PENDING
hbv	VARCHAR(15)	NOT NULL	Hepatitis B surface antigen
hcv	VARCHAR(15)	NOT NULL	Hepatitis C antibody
syphilis	VARCHAR(15)	NOT NULL	VDRL test
malaria	VARCHAR(15)	NOT NULL	Smear / antigen test
technician_name	VARCHAR(80)	NOT NULL	Staff who performed and certified tests
tested_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	Verification timestamp

Table 5: `blood\_requests`

Column Name	Data Type	Constraints	Description
id	INTEGER	PRIMARY KEY AUTOINCREMENT	Requisition ID
hospital_name	VARCHAR(100)	NOT NULL	Name of requesting hospital
patient_name	VARCHAR(100)	NOT NULL	Intended transfusion recipient
blood_group	VARCHAR(5)	NOT NULL	Recipient blood group
component_type	VARCHAR(30)	NOT NULL	Component required

units_requested	INTEGER	NOT NULL	Number of bags required
urgency	VARCHAR(15)	NOT NULL	ROUTINE, URGENT, CRITICAL
doctor_name	VARCHAR(100)	NOT NULL	Attending physician
status	VARCHAR(20)	NOT NULL	PENDING, APPROVED, REJECTED, FULFILLED
created_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	Submission timestamp

Table 6: `blood\_camps`

Column Name	Data Type	Constraints	Description
id	INTEGER	PRIMARY KEY AUTOINCREMENT	Camp ID
camp_name	VARCHAR(100)	NOT NULL	Name/Theme of donation camp
venue	VARCHAR(150)	NOT NULL	Location address / campus
camp_date	DATE	NOT NULL	Date of drive
start_time	VARCHAR(10)	NOT NULL	e.g. "09:00 AM"
end_time	VARCHAR(10)	NOT NULL	e.g. "05:00 PM"
organizer_name	VARCHAR(100)	NOT NULL	Hosting organization (e.g. PESU Rotaract)
organizer_phone	VARCHAR(20)	NOT NULL	Contact helpline

8. ABO / Rh Compatibility Matrix Reference

For reference in automated matching and clinical verification:

Recipient Group	Whole Blood / PRBC Compatible Donor Groups	FFP (Plasma) Compatible Donor Groups
\$O^--\$	\$O^--\$	\$O^-, O^+, A^-, A^+, B^-, B^+, AB^-, AB^+\$ (All)
\$O^+-\$	\$O^-, O^+\$	\$O^+, A^+, B^+, AB^+\$
\$A^--\$	\$O^-, A^--\$	\$A^-, A^+, AB^-, AB^+\$
\$A^+-\$	\$O^-, O^+, A^-, A^+\$	\$A^+, AB^+\$
\$B^--\$	\$O^-, B^--\$	\$B^-, B^+, AB^-, AB^+\$
\$B driver	\$O^-, O^+, B^-, B^+\$	\$B^+, AB^+\$
\$AB^--\$	\$O^-, A^-, B^-, AB^--\$	\$AB^-, AB^+\$
\$AB^+-\$	Universal Recipient (All groups)	\$AB^+\$ only

(Universal Red Cell Donor: \$O^--\$; Universal Plasma Donor: \$AB^+\$ / \$AB^--\$).

End of Software Requirements Specification

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